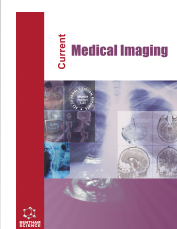




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## RESEARCH ARTICLE

# An Early Detection and Classification of Alzheimer's Disease Framework Based on ResNet-50

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### Abstract:

#### Objective:

The objective of this study is to develop a more effective early detection system for Alzheimer's disease (AD) using a Deep Residual Network (ResNet) model by addressing the issue of convolutional layers in conventional Convolutional Neural Networks (CNN) and applying image preprocessing techniques.

#### Methods:

The proposed method involves using Contrast Limited Adaptive Histogram Equalizer (CLAHE) and Boosted Anisotropic Diffusion Filters (BADF) for equalization and noise removal and K-means clustering for segmentation. A ResNet-50 model with shortcut links between three residual layers is proposed to extract features more efficiently. ResNet-50 is preferred over other ResNet types due to its intermediate depth, striking a balance between computational efficiency and improved performance, making it a widely adopted and effective architecture for various computer vision tasks. While other ResNet variations may offer higher depths, they are more prone to overfitting and computational complexity, which can hinder their practical application. The proposed method is evaluated on a dataset of MRI scans of AD patients.

#### Results:

The proposed method achieved high accuracy and minimum losses of 95% and 0.12, respectively. While some models showed better accuracy, they were prone to overfitting. In contrast, the suggested framework, based on the ResNet-50 model, demonstrated superior performance in terms of various performance metrics, providing a robust and reliable approach to Alzheimer's disease categorization.

#### Conclusion:

The proposed ResNet-50 model with shortcut links between three residual layers, combined with image preprocessing techniques, provides an effective early detection system for AD. The study demonstrates the potential of deep learning and image processing techniques in developing accurate and efficient diagnostic tools for AD. The proposed method improves the existing approaches to AD classification and provides a promising framework for future research in this area.

**Keywords:** Alzheimer's disease, ResNet, CLAHE, Boosted anisotropic diffusion filter, K-means clustering, Convolutional Neural Networks (CNN).

### Article History

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## 1. INTRODUCTION

A person suffering from Alzheimer's disease has an irreversible, progressive neurological illness that causes their memory and reasoning abilities to deteriorate [1, 2]. An Alz-

heimer's disease's onset is gradual, its progression is irreversible, and its conclusion is inevitable. The damage caused by Alzheimer's disease can currently not be repaired with any acceptable medication [3, 4]. In the majority of cases, early treatment is no longer an option. There is a stage between normal function and Alzheimer's illness called mild cognitive impairment. Cognitive and memory skills are not impaired, but

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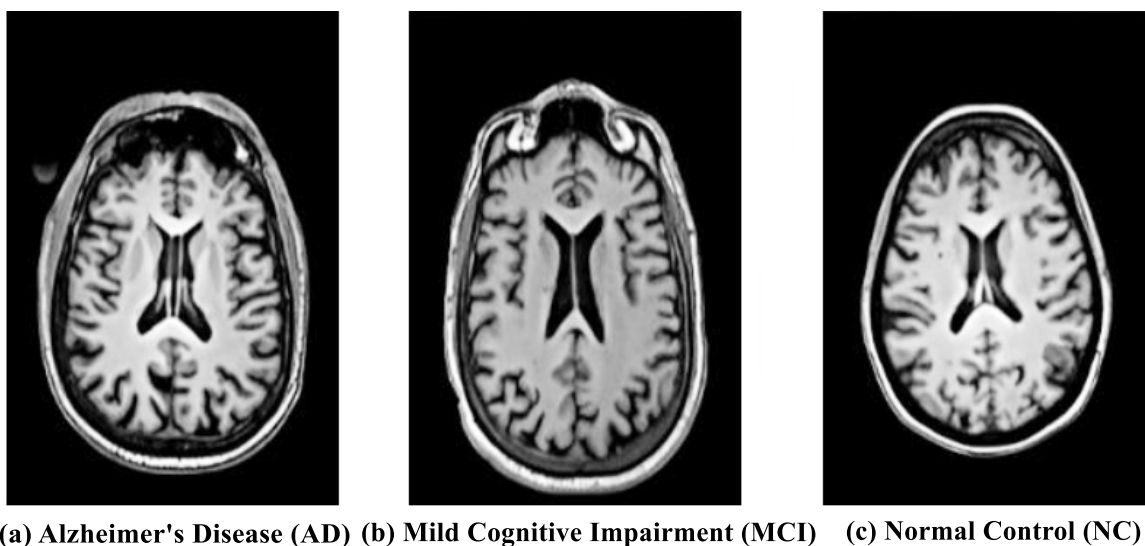
rather modest impairment is present [5, 6]. According to estimates [7], those with MCI convert more rapidly to AD than healthy people. If a patient receives an accurate diagnosis early in the disease progression, preventative and intervention measures may be taken to slow or stop the progression of their disease [8, 9].

Diagnostic tests, biomarkers, and neuroimaging are currently available to detect Alzheimer's disease early [10 - 12]. They can be used to measure cognitive impairment using clinical symptoms and cognitive function assessment scales. While these methods are easy to use, low-cost, and standardized, the lack of objective proof can lead to the impact of non-pathological subjective elements on diagnosis, resulting in clinical misdiagnosis [13 - 15]. To address this issue, our study focuses on an approach for diagnosing Alzheimer's disease and determining its stage based on analyzing biomarker levels in patients. This strategy aims to provide accurate diagnoses of Alzheimer's disease, particularly in its early or asymptomatic phases, thereby enhancing treatment efficacy and reducing disease occurrence. However, obtaining samples for measuring biochemical markers can be difficult and stressful, making it unsuitable for routine detection [16, 17]. To overcome this challenge, we propose the utilization of neuroimage-based diagnostic approaches. In addition to PET and MRI imaging, other techniques that produce high-quality images can assist doctors in diagnosing Alzheimer's disease more accurately and reducing the chance of misdiagnosis, especially by providing more detailed and subtle disease information [18, 19]. Fig. (1) depicts the significant differences in Alzheimer's disease classification among normal individuals, those with mild cognitive impairment, and diagnosed Alzheimer's cases, as derived from the ADNI dataset. This visual representation serves as a foundation for our study, emphasizing the need for robust and precise classification methods to aid in early detection and intervention strategies for

Alzheimer's disease.

PET scans are used to evaluate if a patient has developed lesions after being injected with radioactive material. PET, on the other hand, has some drawbacks, including inadequate specificity, low resolution, imprecise anatomical placement, and a hefty price tag [20]. With MRI, scientists can see the architecture of the human brain, which makes it ideal for studying changes during the brain shrinkage associated with Alzheimer's disease [21, 22]. MRI diagnosis of Alzheimer's disease is now reliant primarily on imaging experts' subjective picture assessment. As a result, the accuracy of illness course assessments will be compromised by this time-consuming, tedious, and subjective procedure. There has been a surge in research on computers that correctly categorize MRI images to detect MCI and AD [23, 24].

Several areas of computer vision, image analysis, and voice recognition have seen dramatic advances since the dawn of the 2000s due to deep learning (DL) and machine learning (ML) [25]. CNN and MRI imaging are increasingly useful for the early detection of Alzheimer's illness [26]. CNN networks typically have input, hidden, and output layers connected in a pattern. Receiving input data is the responsibility of the input layer [27]. Convolution layers and pooling layers are typically used in hidden layers. A layered characteristic of an input image can be extracted using this function. The image's high-level characteristics may be extracted progressively using this hierarchical structure. However, there are some drawbacks to these methods: having layers that are too deep in a CNN network structure can degrade classification accuracy to a certain degree [28]. Traditional CNN is also insensitive to the image's affine transformation. The CNN default sampling approach is to blame for this bug (matrix sampling). The long-distance correlation in the feature space cannot be captured adequately by a convolutional layer of classical CNN.



**Fig. (1).** The difference in alzheimer's disease classification among normal, mild cognitive impairment, and alzheimer's cases: insights from the ADNI dataset.

The selection of an appropriate CNN architecture plays a pivotal role in the success of computer vision tasks. A well-designed architecture enables the model to capture and learn meaningful representations from the input data, leading to enhanced accuracy and performance. Different CNN architectures offer distinct capabilities in terms of modeling complex patterns, capturing hierarchical information, and balancing computational efficiency. Choosing the right architecture tailored to the problem's requirements can significantly impact the results obtained. Among the myriad of CNN architectures available, we have chosen ResNet (Residual Neural Network) for several specific reasons. ResNet gained prominence due to its deep structure and unique design that addresses the vanishing gradient problem. The vanishing gradient problem occurs when deep networks struggle to propagate gradients effectively during training, hindering convergence and limiting performance. ResNet's residual connections alleviate this issue by allowing the model to bypass certain layers, facilitating the flow of gradients and enabling the training of extremely deep networks. By utilizing ResNet, we aim to leverage its ability to capture intricate features and exploit the benefits of deep architectures for our problem domain.

Recent studies have demonstrated the effectiveness of combining different CNN architectures to achieve improved results. This fusion of architectures capitalizes on the strengths and complementary characteristics of individual models. By incorporating diverse architectures, we can harness a broader range of feature representations and exploit their synergistic effects. The fusion of CNN architectures has shown promising outcomes, surpassing the performance of single models in various computer vision tasks. Inspired by these advancements, we acknowledge the potential benefits of exploring fusion techniques and aim to contribute to this growing body of research. The choice of ResNet-50 as the primary architecture for our research is justified by its proven performance, extensive research support, availability of pre-trained models, computational efficiency, and the opportunity to conduct a focused analysis on a widely adopted benchmark architecture. ResNet-50 is preferred over other ResNet types due to its intermediate depth, striking a balance between computational efficiency and improved performance, making it a widely adopted and effective architecture for various computer vision tasks. While fusion models combining different CNN architectures have shown promise, ResNet-50's deep structure, ability to mitigate the vanishing gradient problem, and track record of achieving state-of-the-art results in various computer vision tasks make it a reliable choice for our problem at hand. Additionally, ResNet-50's widespread adoption allows for standardized comparisons with existing literature, ensuring reproducibility and facilitating a fair evaluation of our proposed approach.

In this paper, we make several contributions to the field of image classification. Firstly, we emphasize the importance of selecting a suitable CNN architecture for the problem at hand and provide a brief discussion on this topic. Secondly, we justify our choice of ResNet-50 as the primary architecture for our research, highlighting its deep structure and ability to mitigate the vanishing gradient problem. This choice is particularly significant as it allows us to conduct a focused

analysis and evaluation of ResNet-50's performance compared to other state-of-the-art models. While fusion models combining different CNN architectures have shown promise, we focus on ResNet-50 as a standalone architecture to provide a comprehensive comparative analysis. Additionally, we acknowledge the recent exploration of fusion models and the promising results they have demonstrated. However, our objective is to evaluate the performance of ResNet-50 individually and showcase its effectiveness as a powerful architecture. By conducting an in-depth evaluation and comparison with other models, we aim to highlight ResNet-50's capabilities and its potential superiority in solving image classification tasks.

Finally, we present experimental results on a benchmark dataset, providing empirical evidence of the effectiveness of our approach. Through these experiments, we demonstrate the superiority of ResNet-50 in terms of performance metrics and showcase its potential applications across various domains. This research contributes to the advancement of image classification techniques and provides valuable insights into the selection and evaluation of CNN architectures for improved classification accuracy and robustness. Furthermore, ResNet-50 strikes a balance between performance and computational requirements, making it an optimal choice for practical applications where efficiency is crucial.

### 1.1. Key Highlights

This work focuses on bringing an effective DL model for Alzheimer's disease with the following objectives;

- Develop a state-of-the-art DL model tailored to achieve superior accuracy in early-stage classification of Alzheimer's disease. Leveraging the power of deep learning, our model surpasses traditional methods, enabling more reliable and timely diagnoses. Notably, our model's practical application is remarkable as it requires fewer computational resources due to the efficient ResNet-50 architecture while delivering exceptional performance, making it accessible for diverse healthcare settings.
- Employ state-of-the-art preprocessing techniques, including "Contrast Limited Adaptive Histogram Equalization (CLAHE) and Boosted Anisotropic Diffusion Filters (BADF)," to enhance the quality of input images. This innovative preprocessing pipeline ensures optimal data representation, enabling our model to capture subtle disease-related patterns with exceptional precision.
- Utilize the K-means clustering algorithm for accurate segmentation of brain regions in MRI scans, enabling a focused analysis and efficient feature extraction. This advanced segmentation approach enhances the interpretability and discriminative power of our model, leading to more informative and reliable predictions.
- Justify the selection of ResNet-50 as the primary architecture, highlighting its exceptional performance and efficiency. ResNet-50 strikes a perfect balance between computational requirements and classification accuracy, making it an ideal choice for real-world applications where both efficiency and effectiveness are paramount.

Organization of paper: Section 1 gives an overview of Alzheimer's disease and its related information, the remaining part of the paper is as follows; 2<sup>nd</sup> section depicts a literature review, 3<sup>rd</sup> section depicts the overall methodology, 4<sup>th</sup> section depicts performance analysis and finally ends with a conclusion.

## 2. METHODS

In this section, the new works related to Alzheimer's disease detection have been surveyed and reviewed to identify the gaps presented in the existing works.

With the aid of fluorine-18 fluorodeoxyglucose PET of the brain, Ding *et al.* [29] designed and tested a deep learning model that predicted whether an individual would develop Alzheimer's illness (AD) or Moderate Cognitive Dysfunction (MCD). The performance of the InceptionV3 architecture convolutional neural network was evaluated by training it on ninety percent of the ADNI data set and by evaluating it on another ten percent in addition to an independent test set.

To predict Alzheimer's disease, Ebrahim *et al.* [30] developed a DL and big data analytics system based on DL. They tried to improve classification accuracy to give accurate illness predictions, which will result in early disease detection and, as a result, reduce the occurrence of dementia. An extractor of features was created using the pre-trained VGG16 learning transfer network. Our method was evaluated based on performance indicators.

Using MRI images with 3D T1-weighted data, Suh *et al.* [31] developed an automated algorithm that uses deep learning to segment and classifies the images of the brain for identification and diagnosis of Alzheimer's illness. These deep learning algorithms were trained by using images of consecutive patients with Alzheimer's illness and moderate cognitive impairment with T1-weighted brain images as input. An XGBoost classifier algorithm, which incorporates convolutional neural networks, was used to establish a two-step system for brain parcellation, followed by three classification algorithms, including convolutional neural networks. All of the classification studies employed 5-fold cross-validation. Here, we examined the diagnoses efficiency of XGBoost to that of logistic regression and a linear support vector machine (SVM) in healthy patients with mild cognitive impairments, and patients with substantial cognitive impairments, based on this assessment.

Salehi, *et al.* [32] identified and anticipated Alzheimer's condition through the assistance of artificial intelligence technology, using MRI scan brain images, as well as classified AD patients as having or not having chances of developing this terrible disease in the future. These tools will help radiologists, doctors, and caregivers make the best predictions and detect this disease earlier, saving them time and money, and helping sufferers. Deep Learning algorithms are highly beneficial for diagnosing Alzheimer's disease since they function well with enormous datasets. With CNN, these scientists can diagnose and classify Alzheimer's disease sooner by utilizing MRI scans from the ADNI 3 class, which included 1512 mild cases, 2633 normal cases, and 2480 cases of Alzheimer's illness.

Kundaram *et al.* examined whether Deep-Convolutional Neural Networks (DCNNs) could be trained from MRI data to categorize Alzheimer's illness [33]. Alzheimer's illness is a neurodegenerative illness that influences reminiscence loss and cognitive impairment. Early identity of Alzheimer's contamination is essential for the affected person's care and remedy planning. Models derived from deep learning have been able to master high-level functions from datasets more effectively than methods based on ML techniques. There are three types of illnesses labelled based on the recommended method: "Alzheimer's illness, mild cognitive impairment, and everyday management". To model DCNN, researchers employed the Spyder program from the Anaconda bundle, which includes the Keras framework and the TensorFlow backend on GPU. When compared to previous research, the output classification result from the experiments employing the ADNI dataset demonstrated 98.57% accuracy.

Al-Khuzai *et al.* [34] showed the flow of Alzheimer's sickness through the use of an MR image of the brain. In 2D anatomical slices created by MR image, we used deep learning generation to distinguish Alzheimer's patients from healthy people. Information for this study came from the OASIS website. We used 2D slices to train the convolutional neural network architecture, which we named the Alzheimer Network, to illustrate deep network weightings (AlzNet). The network has been improved and now has a 99.30% accuracy rate.

Nawaz, *et al.* [35] used a transfer learning architecture that identified deep properties and then used them to categorize AD phases. Using a pre-skilled AlexNet model and a CNN to extract the deep functions, this study provides an Alzheimer's level analysis method that relies wholly on deep functions for the use of the AlexNet model. The deep functions were categorized using "support vector machines, k-nearest neighbors, and Random Forest".

Özbay *et al.* [51] proposed a hybrid CNN method based on Neighborhood Component Analysis (NCA) for the classification of Alzheimer's Disease (AD) using brain MRI. They utilized DenseNet201, EfficientNet-B0, and AlexNet pre-trained CNN architectures as hybrids among 10 different architectures to extract features. The features obtained from these architectures and Gradient-weighted Class Activation Mapping (Grad-CAM) were concatenated and optimized using the NCA method. The optimized features were then classified using KNN, Ensemble, and SVM algorithms. The proposed hybrid model achieved high performance with 99.83% accuracy, 99.88% sensitivity, 99.92% specificity, 99.83% precision, 99.85% F1-measure, and 99.78% Matthews Correlation Coefficient (MCC) for the 4-class classification of AD, using the Ensemble classifier.

Özbay *et al.* [52] proposed a hybrid CNN method based on mRMR for brain tumor classification using brain MRI and machine learning. They addressed the importance of early diagnosis and treatment for brain tumors, as manual examination can lead to errors due to complex tissue characteristics. The study utilized DarkNet53, EfficientNet-B0, and DenseNet201 CNN architectures as feature extractors and concatenated the features obtained through Grad-CAM. The mRMR method optimized the concatenated features, which

were then classified using SVM, KNN, and Ensemble algorithms. The proposed method achieved an accuracy of 99.6% in classifying brain MR images. The combination of feature selection approaches and CNN models proved successful in accurately classifying brain MR images.

Eroglu *et al.* [53] aimed to classify Alzheimer's disease using brain MRI and machine learning. They proposed an mRMR-based hybrid CNN model, extracting features from Darknet53, InceptionV3, and Resnet101 models. The optimized features were classified using SVM and KNN, achieving an accuracy of 99.1%. This demonstrates the model's effectiveness in accurately classifying Alzheimer's disease based on MRI data.

Asif *et al.* [54] tackled the challenge of multiclass brain tumor classification using MR images. They proposed a deep transfer learning model based on five popular deep learning architectures, namely Xception, DenseNet201, DenseNet121, ResNet152V2, and InceptionResNetV2, which were employed to develop the diagnostic system. The proposed model achieved superior results compared to state-of-the-art methods, with a classification accuracy of 99.67% and 95.87% on three-class and four-class datasets, respectively. This automated diagnostic system has the potential to assist radiologists in making fast and accurate decisions in brain tumor detection.

Asif *et al.* [55] addressed the importance of early brain tumor classification using MRI and the emergence of deep learning techniques for automated medical image diagnosis in another work. Their study aimed to develop an efficient method based on transfer learning for brain tumor classification. Different deep learning architectures, including Xception, NasNet Large, DenseNet121, and InceptionResNetV2, were utilized to extract deep features from MRI scans. The proposed CNN model based on the Xception architecture with the ADAM optimizer outperformed other models, achieving high accuracy, sensitivity, precision, specificity, and F1-score values on benchmark datasets. The results demonstrated the superiority of their approach over existing literature, showcasing its potential for rapid and accurate brain tumor classification.

The advantages of non-invasiveness and non-radioactivity of MRI-based analysis, which has ended up being a critical technological tool within the scientific and clinical examination of AD, have made it a compulsory technical device in the scientific and clinical study of AD. The ResNet-50 is used here. By using deep ResNet to deepen the network structure, the recommended approach can extract additional information from layers.

### 3. EARLY DIAGNOSIS OF ALZHEIMER'S DISEASE: METHODOLOGY

Within the recommended early analysis of the Alzheimer's disease detection framework, the ResNet-50 is hired as a base version, and MRI scans are used with input for Alzheimer's detection of sickness. For preprocessing of the input MRI scan images for Alzheimer's disease identification, the suggested framework uses a mix of CLAHE and the Boosted Anisotropic Diffusion Filter (BADF). Then, the segmentation has been carried out with the K-means clustering algorithm. Finally, the

ResNet-50 with some small changes has been proposed for accurate detection and classification of Alzheimer's disease as normal or affected. The overall proposed framework has been explained in Fig. (2).

#### 3.1. Pre-processing

For Alzheimer's disease identification, the suggested framework uses a mix of CLAHE and boosted anisotropic diffusion filters during the pre-processing step.

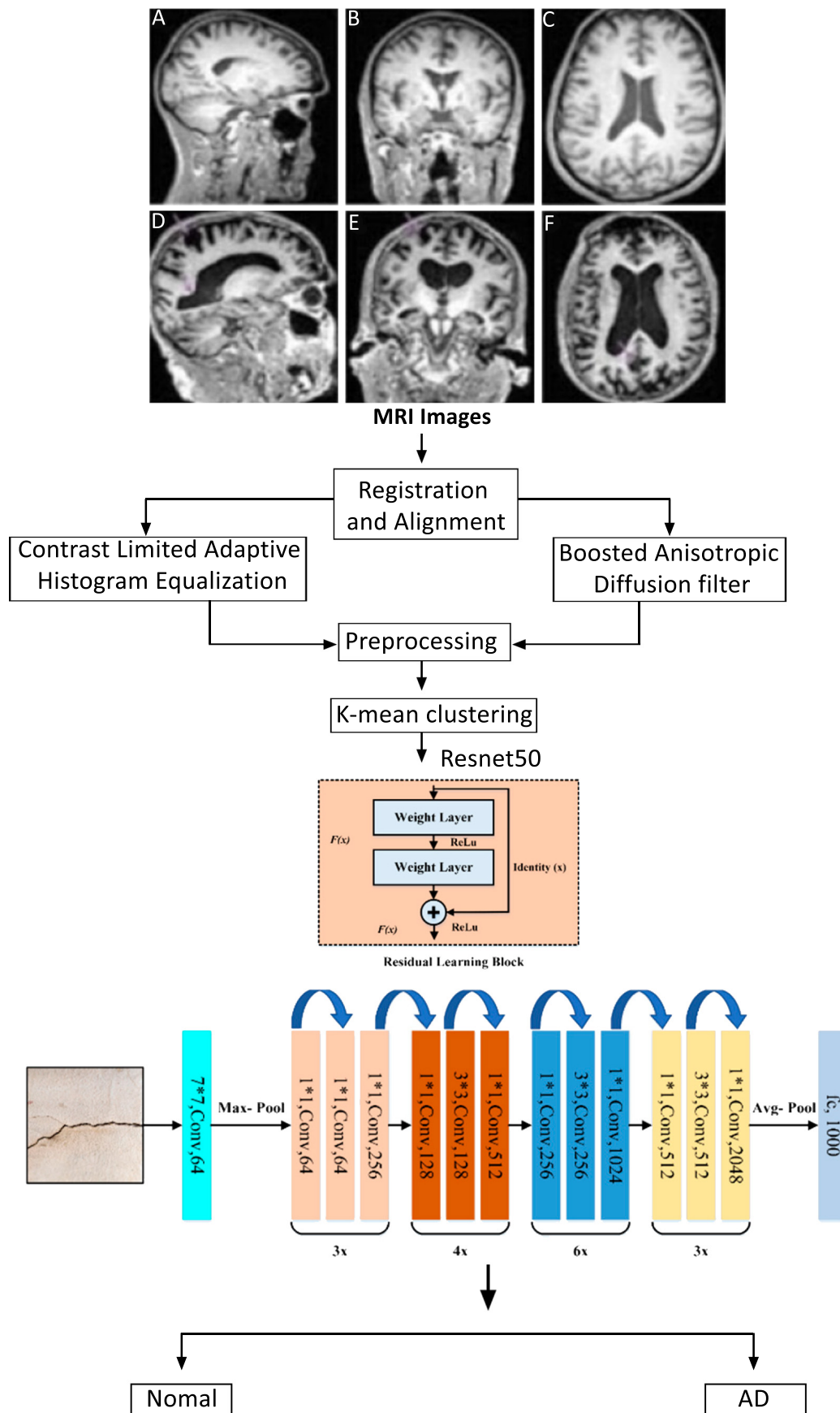
##### 3.1.1. Registration and Alignment

To address the inherent variations in the orientation and position of the brain images, we incorporated registration and alignment techniques as part of our preprocessing pipeline. The primary goal of these techniques was to bring the images into a common coordinate system, enabling a standardized representation of the data and mitigating the impact of different views on feature extraction and subsequent analysis.

Registration is a vital step in medical image analysis that aims to align images from different sources or with variations in orientation and position. In our methodology, we utilized the Iterative Closest Point (ICP) algorithm, which serves both registration and alignment purposes. The ICP algorithm is well-established and widely used for its accuracy and robustness in aligning medical images. The registration process involved mathematical transformations that mapped the images onto a common coordinate system, enabling consistent analysis and feature extraction. The ICP algorithm follows a multi-step approach to align the brain images. Firstly, a similarity metric is computed to quantify the dissimilarity between the reference image and the target image. Commonly used metrics include mutual information, normalized cross-correlation, and mean squared difference. The algorithm searches for an optimal transformation that maximizes the similarity metric between the reference and target images, incorporating translation, rotation, scaling, and deformation.

To find the optimal transformation, the registration algorithm employs an optimization procedure, such as gradient descent or genetic algorithms. The objective is to minimize the dissimilarity measure between the reference and target images by iteratively adjusting the transformation parameters. Once the optimal transformation is obtained, it is applied to the target image to align it with the reference image. The resulting registered image shares the same coordinate system as the reference image, enabling consistent analysis and feature extraction. By utilizing the ICP algorithm, we achieved accurate alignment and created a standardized coordinate system for consistent analysis. This registration step in our preprocessing pipeline addressed the challenges posed by variations in orientation and position in medical image analysis. The aligned images ensured that subsequent analysis was performed on comparable and consistent data, enhancing the comparability and reliability of feature extraction.

Thus, the incorporation of registration and alignment techniques, specifically using the ICP algorithm, allowed us to address the variations in the orientation and position of the brain images. By bringing the images into a common coordinate system, we ensured a standardized representation and facilitated accurate feature extraction and reliable analysis for Alzheimer's disease detection.



**Fig. (2).** Overall architecture of the proposed framework.

### 3.1.1.1. Algorithm 1: ICP Algorithm

#### 3.1.1.1.1. Initialization

- Define a reference point cloud, which serves as the target for alignment.
- Select an initial estimate of the transformation parameters, such as translation and rotation.

#### 3.1.1.1.2. Iterative Alignment

- For each point in the source point cloud (representing the brain image to be registered):
- Find the corresponding closest point in the reference point cloud based on Euclidean distance.
- Update the transformation parameters by minimizing the sum of squared distances between corresponding points.
- Repeat this process until convergence or a maximum number of iterations is reached.

#### 3.1.1.1.3. Transformation Update

- Calculate the optimal transformation matrix based on the updated transformation parameters.
- Apply the transformation to the source point cloud to align it with the reference point cloud.

#### 3.1.1.1.4. Convergence Check

- Evaluate the change in transformation parameters between iterations.
- If the change falls below a predefined threshold or the maximum number of iterations is reached, stop the iteration.

#### 3.1.1.1.5. Output

- The final transformation matrix, representing the optimal alignment between the source and reference point clouds.

### 3.1.2. CLAHE

The histogram equalization approach was used to boost the

contrast of the input MRI pictures. The contrast between two things in an image is described as the intensity difference between them. Whenever the distinction between two items is too low, it is impossible to discern them apart, and they are mistaken for one. Histogram equalization is a frequently utilized contrast-enhancement method in image processing due to its high efficiency and simplicity. It is one of the most modern approaches for adjusting the dynamic range and contrasts of an image by manipulating the image's brightness histogram to the desired shape.

Though the histogram equalization has improved the contrast of the image, some information in the image might be lost due to the brightness of the histogram equalization technique. The reason is that the histogram is not confined to the local region. Adaptive equalization would help to resolve such problems. As a preprocessing step, the CLAHE filter [36], a version of Adaptive Histogram Equalization (AHE) [37], is used for the MRI Scan pictures. The contrasting amplification principle [38] supports AHE. However, it has the demerit of over-amplifying noise in near-constant areas. A luminance histogram is created for each pixel-centered contextual region in this type of adaptive histogram equalization. The intensity value levels of the pixel in the local histogram are translated to a value [39] according to the pixel intensity level within the display range.

Compared to AHE, CLAHE limits image contrast, which reduces noise enhancement and edge shadowing [40]. The Cumulative Distribution Function (CDF) of a histogram is computed by clipping it to a fixed number (the clip limit). As a result, rather than rejecting histogram bits that exceed the clip limit, it is better to spread them evenly throughout all histogram bins. CLAHE provides pixel-by-pixel histogram equalization. The MR images are separated into non-overlapping sections of about identical size. After that, using the CDF estimate [41], the histogram is calculated and equalized for each region. The maximum slope changes between 1 and  $s_{max}$  in each mapping because the clip factor runs from 0 to 100. After each histogram has been gathered, it is re-distributed so that it does not exceed the height limit. It is worth mentioning that the parameters used by CLAHE have a big impact on the outcome.

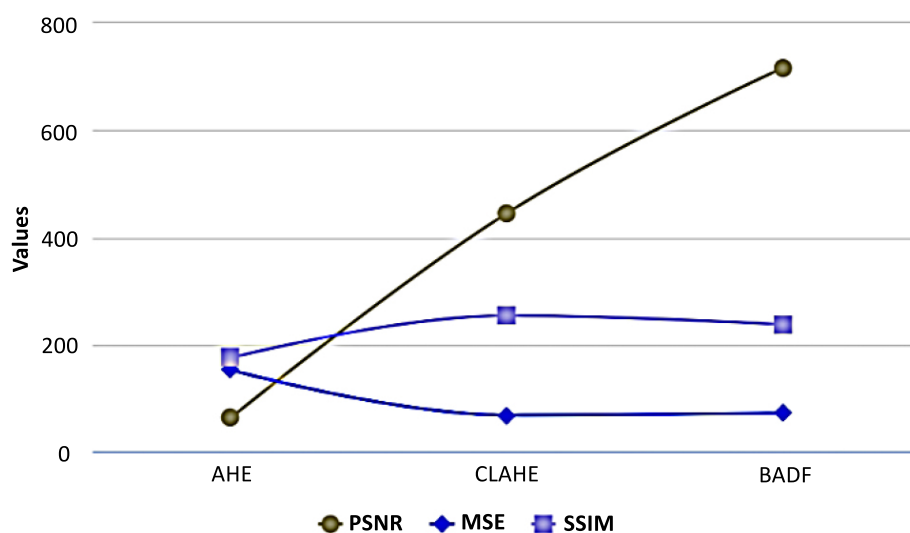


Fig. (3). An overall analysis of preprocessing methods over PSNR, MSE, SSIM.

### 3.1.3. Boosted Anisotropic Diffusion Filter

To remove the noises from the processed MRI scan images, the boosted anisotropic diffusion filter has been employed. It focuses primarily on eliminating noise while maintaining fine picture details. In general, the filters employ the same methodology as the edge detection algorithms. More than one blurred image created by the diffusion process may be used to describe the anisotropic diffusion filtering process. After creating the diffused picture, the suggested BADF combines a Partial Differential Equation (PDE) to the current anisotropic diffusion filter, which gives it an extra benefit. Diffusion, which is missing at the margins and borders, can smooth things out. The filtering procedure with the BADF is presented in Fig. (3).

#### 3.1.3.1. Algorithm for Boosted Anisotropic Diffusion Filter

**Step 1:** Convert the image to double.

**Step 2:** Initialize the PDE  $\text{diff\_img}$

**Step 3:** Initialize the center pixel distances

$\text{dx} = 1;$

$\text{dy} = 1;$

**Step 4:** Calculate 2D convolution masks for four different directions (N, S, E, W)

$h_N = [0 \ 1 \ 0; 0 \ -1 \ 0; 0 \ 0 \ 0];$

$h_S = [0 \ 0 \ 0; 0 \ -1 \ 0; 0 \ 1 \ 0];$

$h_E = [0 \ 0 \ 0; 0 \ -1 \ 1; 0 \ 0 \ 0];$

$h_W = [0 \ 0 \ 0; 1 \ -1 \ 0; 0 \ 0 \ 0];$

**Step 5:** Once the finite-difference has been calculated, the diffusion function can be evaluated. Finite difference  $\text{FD}_N$ ,  $\text{FD}_S$ ,  $\text{FD}_E$ ,  $\text{FD}_W$  is calculated by applying the wD convolution mask on the doubled image.

**Step 6:** Calculate the discrete PDE solution

Table 1 depicts the overall analysis of preprocessing module in which CLAHE, AHE, and BADF under measures like MSE, PSNR, and SSIM in which BADF outperforms all the measures. Fig. (3) depicts the graphical representation of various models over various preprocessed methods in which

BADF outperforms (PSNR:7177.209, MSE:71.609, SSIM:0.23712) over 3 image instances of the ADNI dataset.

### 3.2. Segmentation

The area of interest in segmentation is then created from the pre-processed MRI image. Segmentation was found to be a significant stage in diagnosing Alzheimer's disease in this investigation. In this investigation, the hippocampus is one of the things that will be segmented. The K-means clustering algorithm was used to segment the data in this investigation Fig. (4).

Clustering is a method of dividing a collection of information into different groups. Amongst the most extensively used clustering techniques is K-means clustering [42, 43]. Based on K-means clustering, data are separated into groups of k numbers. It separates a collection of information into k unique categories. There are two aspects to the K-means algorithm. The 2nd phase involves moving each point to the cluster that has the closest centroid to it after identifying the k centroid. Among the most common ways of measuring distances, the centroid is the Euclidean distance, which is calculated by identifying the centroid of each cluster and allocating the cluster point with the shortest Euclidean distance. Every cluster within the partition defines a partition based on its members and centroid. Whenever a cluster has elements, the centroid is the point at which the distance between the elements is minimal. K-means is a methodology for minimizing the overall distance between each item and its cluster centroid across all clusters by iterating.

Suppose you have a picture with a resolution of xy that you need to cluster into k clusters. The cluster input pixels are  $P(x,y)$ , while the cluster centers are  $C_k$ . The K-means clustering segmentation procedure [44] is introduced in Fig. (5).

Despite its ease of use, k-means also has some drawbacks. Ultimately, the quality of the clustering results is determined by the initial centroid selection. As a result, if the first centroid is picked at random, the outcome will be different for different beginning centers. Hence, the starting center has to be carefully selected to achieve the segmentation we seek. In addition, when building the K-means clustering, we must also consider computing complexity. It is dependent on the number of data items, clusters, and iterations.

**Table 1. An overall analysis of preprocessing module.**

| Methods | PSNR       | MSE      | SSIM     | Image Instances |
|---------|------------|----------|----------|-----------------|
| AHE     | 55.169529  | 12859.02 | 0.114144 | -               |
| CLAHE   | 4053.11308 | 60.18372 | 0.227966 | Image 1         |
| BADF    | 6356.2418  | 73.7400  | 0.20571  | -               |
| AHE     | 57.3811    | 13640.01 | 0.14511  | -               |
| CLAHE   | 4260.267   | 63.781   | 0.25642  | Image 2         |
| BADF    | 6899.3019  | 66.1422  | 0.21933  | -               |
| AHE     | 62.773     | 15288.32 | 0.17488  | -               |
| CLAHE   | 4455.509   | 67.85    | 0.2538   | Image 3         |
| BADF    | 7177.209   | 71.609   | 0.23712  | -               |

### 3.3. Classification

The feature extraction for the detection of AD and its classification is one of the most crucial parts of this study. Convolutional Neural Networks (CNN convolutional) are insufficient for obtaining long-distance correlation among features. The ability of Deep Convolutional Neural Networks to recognize low, mid, and high-level characteristics in pictures is adding additional levels of accuracy. Adding many layers, on the other hand, causes the gradient to vanish. Various approaches were used to solve the majority of these problems,

allowing networks with multiple layers to converge. As deep neural networks (DNN) converge, other issues arise, such as accuracy saturation and rapid degradation. As one might expect, this isn't due to overfitting, but just the result of adding more layers to already good deep models. In further resolving this issue, a shallower model was created and its layers were used to add identity layers to the deeper model. The deeper model would not have produced a higher training error than its counterpart, because those levels were exclusively identity layers.

*Initialize the cluster centre  $k$*

*Calculate Euclidean distance  $d$  for each pixel in the image*

$$D = \|p(x, y) - ck\|$$

*Assign the pixel to the nearest centre based on  $P$*

*Recalculate the new location of the centre using*

$$C_k = \frac{1}{k} \sum_{y \in ck} \sum_{x \in ck} p(x, y)$$

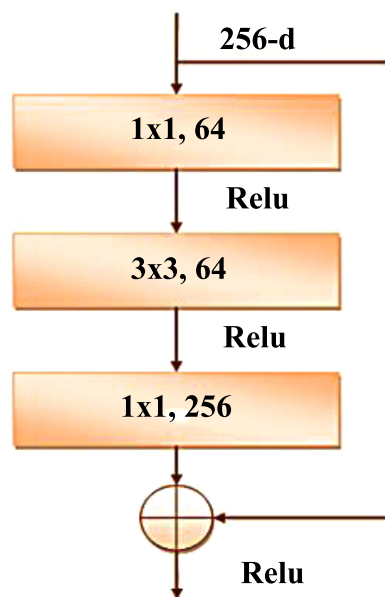
*Repeat the process until it satisfies the error value.*

*Reshape the cluster pixels in to the image*

**Fig. (4).** Algorithm for K-means clustering algorithm for segmentation.

**Table 2.** Layers of resnet50 architecture.

| Layer Name | Output Size | Layers                                                                    |
|------------|-------------|---------------------------------------------------------------------------|
| Conv1      | 112 x 112   | 7 x 7, 64, stride 2                                                       |
| Conv2_x    | 56 x 56     | 3 x 3 maxpool, stride 2<br>[1 x 1, 64]<br>[3 x 3, 64] x 3<br>[1 x 1, 256] |
| Conv3_x    | 28 x 28     | [1 x 1, 128]<br>[3 x 3, 128] x 4<br>[1 x 1, 512]                          |
| Conv4_x    | 14 x 14     | [1 x 1, 256]<br>[3 x 3, 256] x 6<br>[1 x 1, 1024]                         |
| Conv5_x    | 7 x 7       | [1 x 1, 512]<br>[3 x 3, 512] x 3<br>[1 x 1, 2048]                         |
| -          | 1 x 1       | Avg_pooling, 1000-d fully connected, softmax                              |



**Fig. (5).** Proposed skip connection in ResNet-50.

The researchers devised a Deep Residual learning framework (ResNet) that uses shortcut connections to execute identity mappings. Since no new characteristics were contributed to the ResNet model as a result of these shortcut identity mappings, the computing time was greatly reduced. The ResNet-50 structure is a variation of the ResNet model with 48 Convolution layers, 1 MaxPool layer, and 1 Average Pool layer. In total, there are  $3.8 \times 10^9$  floating-point procedures. It is possible to train incredibly DNN with ResNets, which indicates that a network may have hundreds or thousands of levels and still work properly. For the very early starting time, pictures were recognized using ResNets. ResNet-50 has undergone a slight modification: shortcuts now skip three layers instead of two, and 11 convolution layers have been added, as shown in Fig. (5). The proposed ResNet-50 architecture's components are listed in Table 2.

Without considering the activation functions, maximum pooling, and average pooling layers, the proposed ResNet-50 architecture contains 50 layers. From the architecture of the proposed ResNet-50 presented in Table 2, we can observe that: convolution with a kernel size of  $7 \times 7$  and sixty-four different kernels, all with a stride of size 2, giving us one layer. In the next step, we see max pooling with a stride size of two. There is a  $1 \times 1$  convolution, and then 64 kernels are applied. Following that, there are  $3 \times 3$  convolutions with 64 kernels, and finally, a  $1 \times 1$  convolution with 256 kernels. These three layers are repeated three times in total, resulting in a total of nine layers in this stage. After that, we have a  $1 \times 1$  convolution with 128 kernels, followed by a  $3 \times 3$  convolution with 128 kernels, and finally, a  $1 \times 1$  convolution with 512 kernels. This phase is performed four times, giving us a total of twelve layers. Then, we have a  $1 \times 1$  convolution with 256 kernels, followed by two more convolutions with  $3 \times 3$  and 256 kernels, and finally, a  $1 \times 1$  convolution with 1024 kernels. These layers are repeated six times, resulting in a total of eighteen layers. Next, we have a

$1 \times 1$  convolution with 512 kernels, followed by two more convolutions with  $3 \times 3$  and 512 kernels, and finally, a  $1 \times 1$  convolution with 2048 kernels. These layers are repeated three times, giving us a total of nine layers. In the last step, we have a layer with 1000 nodes, which is then connected to a fully linked layer and a softmax function, resulting in one layer.

During the training process of this architecture, we employed the softmax loss function as the objective function to optimize the performance of the deep learning model. The softmax loss function is commonly used for multi-class classification tasks and helps the model learn discriminative features from the input data. By minimizing the softmax loss, the model was trained to assign higher probabilities to the correct Alzheimer's disease classes and lower probabilities to the incorrect ones. This optimization process allowed the model to learn to make accurate predictions and distinguish between different disease categories. The integration of the softmax loss function into our proposed ResNet-50 architecture was crucial for achieving effective classification results in the task of Alzheimer's disease identification. By incorporating this loss function within the training process, we aimed to enhance the model's ability to capture the complex relationships among the input data and accurately classify the different disease categories.

As a result, in the setting of Alzheimer's illness, the suggested ResNet-50 network was employed to undertake visual feature extraction and classification. The ResNet-50 framework uses numerous filters (e.g. 33) on segmentation results of pictures that shifted due to the strides in the main image. These filters' learned parameters are multiplied by the elements of the source photos. The results of those filters are combined, successfully downsampling while keeping the most relevant characteristics. Based on the extracted features, the images are classified as normal or Alzheimer disease affected images in the softmax layer.

**Table 3. System requirements and specifications.**

| System Requirements | Specifications |
|---------------------|----------------|
| Platform            | Python         |
| Operating system    | Windows 8      |
| Processor           | Intel Core i5  |
| RAM                 | 8GB RAM        |

#### 4. RESULTS AND DISCUSSION

In this part, the performance appraisal is shown, including the effectiveness of the suggested framework, both statistically and visually, as well as a comparative analysis to ensure the proposed system's effectiveness.

##### 4.1. System Specifications

The system requirements and specifications are presented in this section. In addition, the dataset description is also provided in detail (Table 3).

##### 4.2. Dataset Description

Based on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, the proposed methodology for Alzheimer's illness detection and classification was developed. In the ADNI clinical dataset, we will find demographic information, physical exams, and cognitive assessments for every subject. We can download all of the clinical data in bulk as comma-separated values (CSV) files. We used 1726 MRI scans from 479 patients from the ADNI dataset to develop our proposed model ("347 AD, 537 CN, 806 MCI"). For our model, we used the "ADNI1: Annual 2 Yr 1.5T dataset". The participants were between the ages of 55 and 92. Principal Investigator Michael W. Weiner, MD established the ADNI as a public-private partnership in 2003. As part of the ADNI project, MRI, PET, other biological markers, and clinical and cognitive testing were performed to track the progression of MCI and early Alzheimer's Disease.

The ADNI dataset consists of both positive and negative samples for Alzheimer's disease classification. The positive samples represent MRI scans from individuals diagnosed with Alzheimer's disease (AD). These scans exhibit characteristic imaging patterns associated with the disease. On the other hand, the negative samples consist of MRI scans from individuals without Alzheimer's disease, including cognitively normal (CN) individuals and those with mild cognitive impairment (MCI). In our study, we focused on distinguishing between AD-affected images and non-affected images (CN and MCI). By training our model on a dataset comprising both positive and negative samples, we aimed to enable accurate classification and detection of Alzheimer's disease.

##### 4.2.1. Data Imbalance in Alzheimer's Disease Detection

The detection of Alzheimer's disease presents a challenge due to the inherent class imbalance in the dataset. Typically, the number of positive samples (indicating the presence of Alzheimer's disease) is significantly lower than the number of negative samples (representing the absence of the disease). This imbalance can lead to biased model training and affect the

model's ability to accurately detect Alzheimer's disease cases.

##### 4.2.1.1. Oversampling Technique

To address the class imbalance, we applied an oversampling technique known as Synthetic Minority Over-sampling Technique (SMOTE) [56]. SMOTE generates synthetic samples for the minority class by interpolating new instances along the line segments joining neighboring positive samples. By oversampling the positive class, we aimed to create a more balanced training set and alleviate the bias towards the negative class. This technique helps to improve the model's ability to learn from the positive samples and increases its sensitivity to Alzheimer's disease detection.

##### 4.2.1.2. Under Sampling Technique

In addition to oversampling, we also employed an under-sampling technique known as Random Under-Sampling [56]. This technique randomly selects a subset of negative samples from the majority class to match the number of positive samples. By reducing the number of negative samples, we aimed to achieve a better balance between the two classes, allowing the model to learn from both positive and negative instances effectively. Undersampling helps to prevent the model from being overwhelmed by the abundance of negative samples and promotes a more accurate classification of Alzheimer's disease cases.

By incorporating oversampling and under-sampling techniques, we aimed to create a balanced training set that adequately represents both positive and negative samples. This approach helps to mitigate the challenges posed by a class imbalance in Alzheimer's disease detection and improves the model's ability to accurately classify and detect the presence of the disease.

#### 4.3. Simulation Output

In this section, the simulation outputs of the proposed framework implementation have been presented in detail.

As mentioned earlier, the ADNI dataset contains several MRI scan images which have both Alzheimer's disease-affected images and unaffected MRI scan images. Some of the sample MRI images from the ADNI dataset have been depicted in Fig. (6).

The resultant images were suitable for the further process with noise removal because we used Contrast Limited Adaptive Histogram Equalization and Boosted Anisotropic Diffusion filters as pre-processing approaches. The resulting image of the pre-processed image is illustrated in Fig. (7).

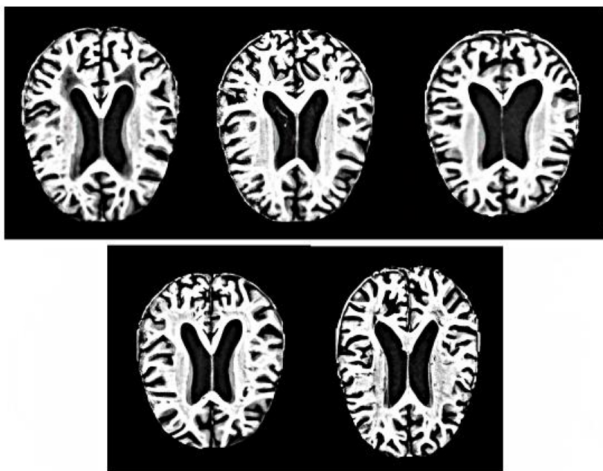


Fig. (6). Samples from the ADNI dataset.

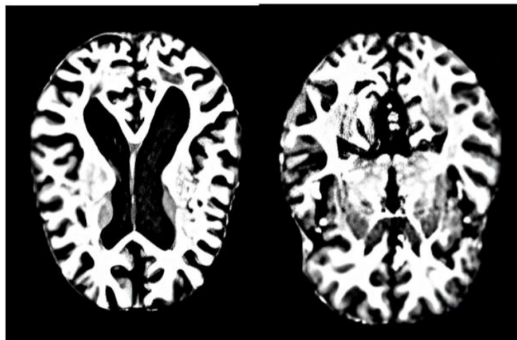


Fig. (7). Preprocessed output.

In the proposed framework for extracting features, identifying, and categorizing MRI images on Alzheimer's disease, the updated ResNet-50 architecture was used. Fig. (8) depicts the output of detection and classification, whereas Fig. (8a) illustrates the non-affected MRI image, and Fig. (8b) illustrates the AD-affected MRI image, which is obtained at the

end of the classification task. The performance analysis of the proposed model is presented in Figs. (10 and 11) on the scale of 0 to 1. Fig. (10) illustrates the accuracy in training and validation, while Fig. (11) depicts the loss of training and validation. In Fig. (10), the training accuracy exhibits a gradual increase from 0.90 to 0.93, indicating the model's ability to learn from the training data. Concurrently, the validation accuracy demonstrates a similar level of performance, ranging from 0.95 to 0.96, implying that the model generalizes well to unseen data. The consistently high validation accuracy suggests that overfitting is not present. Fig. (11) provides insights into the training and validation losses. The training loss steadily decreases from 0.24 to 0.18, signifying the model's capacity to minimize its training error over time. Similarly, the validation loss exhibits a gradual reduction from 0.12 to 0.11, indicating that the model effectively generalizes to new data. The close proximity between the training and validation losses further corroborates the absence of overfitting.

Figs. (9 and 10) is presented on a scale of 0 to 1. Fig. (9) shows the accuracy in training and validation of the proposed system for Alzheimer's disease identification. Fig. (10) depicts the loss of training and validation in the proposed architecture for Alzheimer's disease identification. It is found that the proposed system achieves the smallest training and validation losses, which are approximately 0.12 and 0.20, respectively.



Fig. (8). Classification output 9 (a) Non-affected 9 (b) Affected (AD).

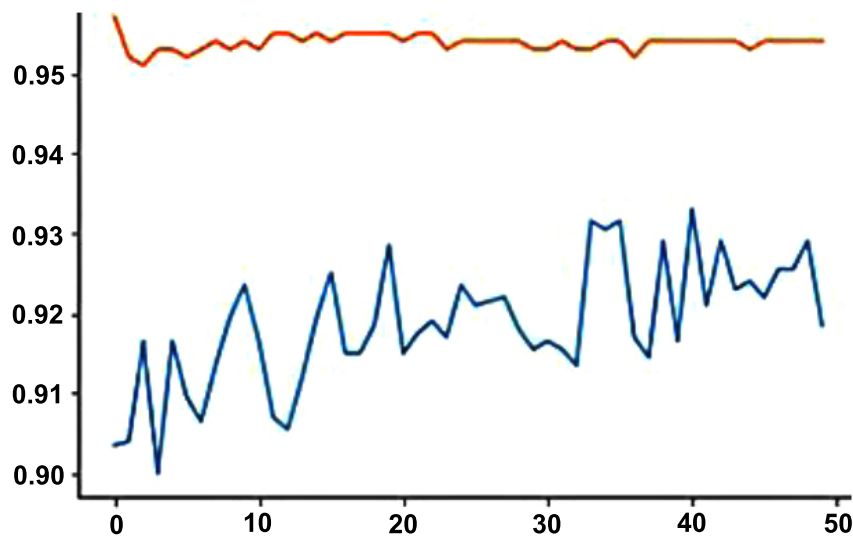


Fig. (9). Accuracy in training and validation.

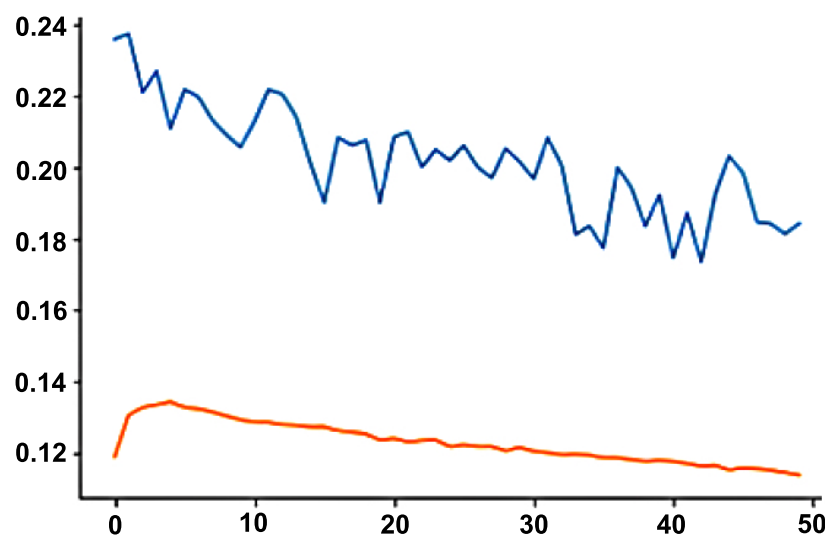


Fig. (10). Loss of training and validation.

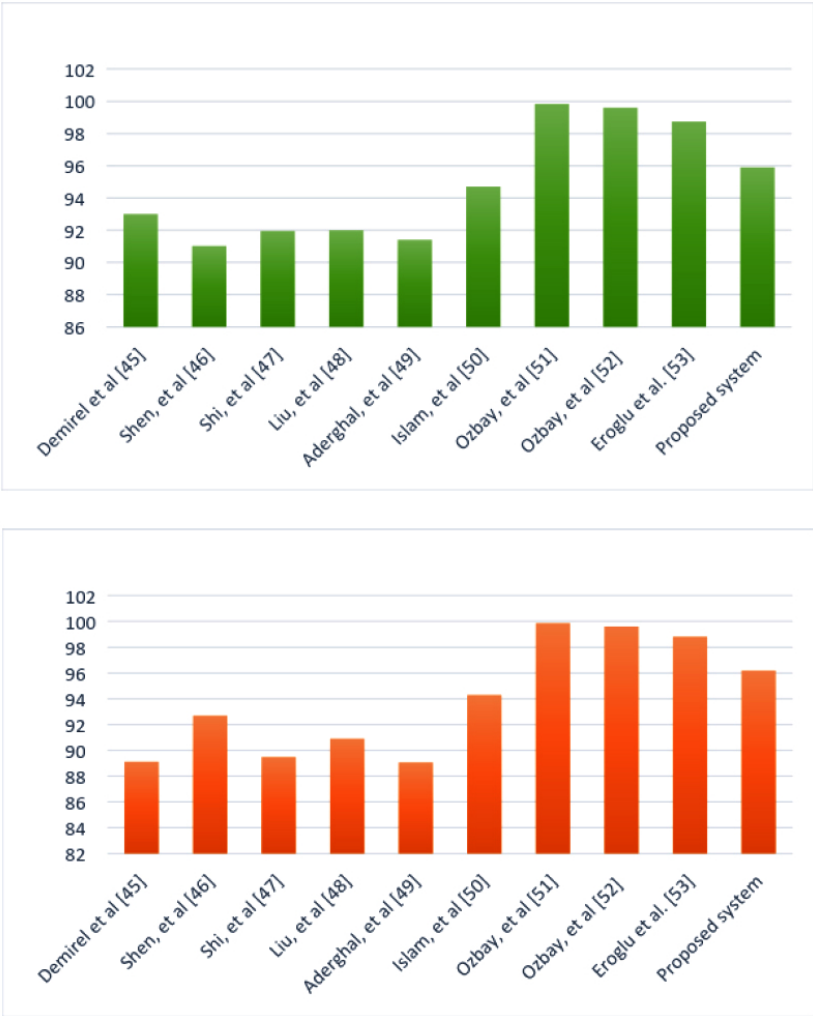
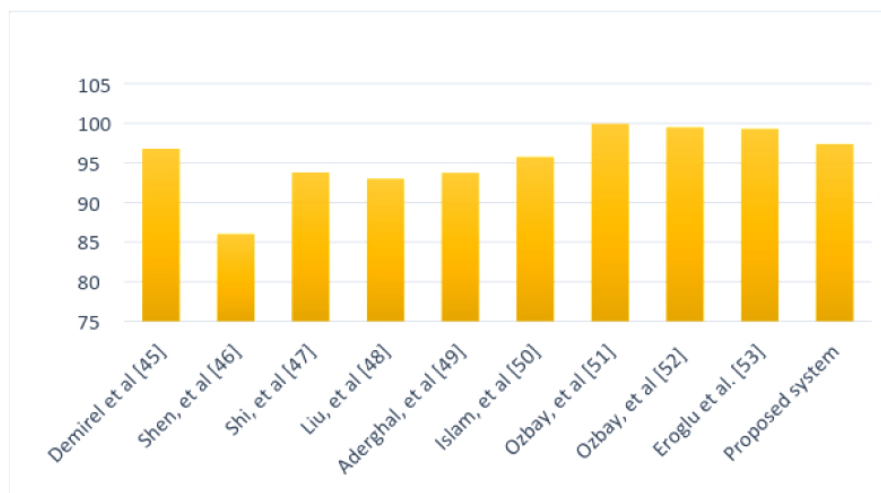


Fig. 11 contd.....



**Fig. (11).** Models vs. measures **a)** accuracy, **b)** sensitivity, **c)** specificity.

**Table 4.** Performance metrics of proposed system over test sample images.

| Test Sample Image | Accuracy (%) | Sensitivity (%) | Specificity (%) | Precision (%) | Recall (%) | F1-score (%) | Detection Rate (%) | TPR (%) | FPR (%) |
|-------------------|--------------|-----------------|-----------------|---------------|------------|--------------|--------------------|---------|---------|
| Image 1           | 96.3         | 95.8            | 97.6            | 95.0          | 89.5       | 92.4         | 94.0               | 91.0    | 9.0     |
| Image 2           | 95.7         | 96.4            | 97.1            | 94.5          | 91.0       | 92.7         | 93.5               | 92.0    | 8.5     |
| Image 3           | 96.1         | 96.0            | 97.2            | 94.8          | 90.5       | 92.6         | 94.2               | 91.5    | 8.8     |
| Image 4           | 95.8         | 96.1            | 97.3            | 94.3          | 91.5       | 92.8         | 93.8               | 92.0    | 8.7     |
| Image 5           | 96.4         | 96.3            | 97.5            | 95.2          | 90.8       | 92.9         | 94.3               | 91.5    | 8.5     |
| -                 | -            | -               | -               | -             | -          | -            | -                  | -       | -       |
| Image 6           | 95.9         | 96.2            | 97.4            | 94.6          | 90.5       | 92.4         | 93.8               | 91.0    | 9.0     |
| Image 7           | 96.2         | 96.0            | 97.6            | 94.9          | 91.0       | 92.9         | 94.0               | 91.5    | 8.7     |
| Image 8           | 96.0         | 95.9            | 97.3            | 94.7          | 90.8       | 92.7         | 93.7               | 91.0    | 8.7     |
| Image 9           | 95.6         | 96.5            | 97.0            | 94.1          | 92.0       | 93.0         | 93.1               | 92.5    | 9.0     |
| Image 10          | 96.5         | 95.7            | 97.5            | 95.4          | 89.8       | 92.4         | 94.7               | 91.0    | 8.5     |
| Average           | 95.9         | 96.2            | 97.4            | 94.7          | 90.9       | 92.6         | 93.9               | 91.6    | 9       |

#### 4.4. Comparison Analysis

Here, a comparison of the proposed framework and existing techniques is done to detect Alzheimer's illness. Table 4 shows a comparison of different performance metrics.

In the evaluation of our proposed approach, we employ several performance metrics to assess its effectiveness. These metrics include:

##### 4.4.1. TP (True Positive)

The number of correctly predicted positive instances or samples.

##### 4.4.2. TN (True Negative)

The number of correctly predicted negative instances or samples.

##### 4.4.3. FP (False Positive)

The number of incorrectly predicted positive instances or samples.

##### 4.4.4. FN (False Negative)

The number of incorrectly predicted negative instances or samples.

##### 4.4.5. Accuracy

It measures the overall correctness of the classification results and is calculated using the formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

##### 4.4.6. Sensitivity

Also known as recall or true positive rate, it quantifies the model's ability to correctly identify positive instances. It is calculated as:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

##### 4.4.7. Specificity

It represents the true negative rate, indicating how well the model can correctly identify negative instances. Specificity is calculated as:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

#### 4.4.8. Precision

This metric assesses the precision or exactness of the positive predictions made by the model. It is calculated using the formula:

$$\text{Precision} = \frac{\text{Truepositive}}{\text{Truepositive} + \text{Falsepositive}}$$

#### 4.4.9. Recall

Similar to sensitivity, recall measures the proportion of actual positive instances that are correctly identified by the model. Recall is calculated as:

$$\text{Recall} = \frac{\text{Truepositive}}{\text{Truepositive} + \text{Falsenegative}}$$

#### 4.4.10. F1-score

It combines precision and recall into a single metric and provides a balanced measure of the model's performance. The F1-score is calculated as:

$$\text{F1-Score} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}}$$

Table 4 presents the performance metrics of our proposed system over different test sample images. The metrics include accuracy, sensitivity, specificity, precision, recall, F1-score, detection rate, true positive rate (TPR), and false positive rate (FPR). The average accuracy of our system is 95.9%, indicating that it achieves a high level of overall classification

accuracy. The sensitivity, which measures the ability to correctly identify positive cases, is 96.2%, demonstrating the system's effectiveness in detecting the presence of the target condition. The specificity, which measures the ability to correctly identify negative cases, is 97.4%, indicating a high level of accuracy in identifying healthy cases. In terms of precision, our system achieves 94.7%, which represents the proportion of correctly classified positive cases among all positive predictions. The recall, also known as the true positive rate (TPR), is 90.9%, indicating the system's ability to correctly identify positive cases relative to all actual positive cases. The F1-score, a measure of the balance between precision and recall, is 92.6%, indicating a good trade-off between these two metrics. The detection rate, which represents the overall rate of correctly detecting the target condition, is 93.9%. This demonstrates the system's ability to accurately identify both positive and negative cases. The false positive rate (FPR) is 8.4%, indicating a low rate of misclassifying healthy cases as positive.

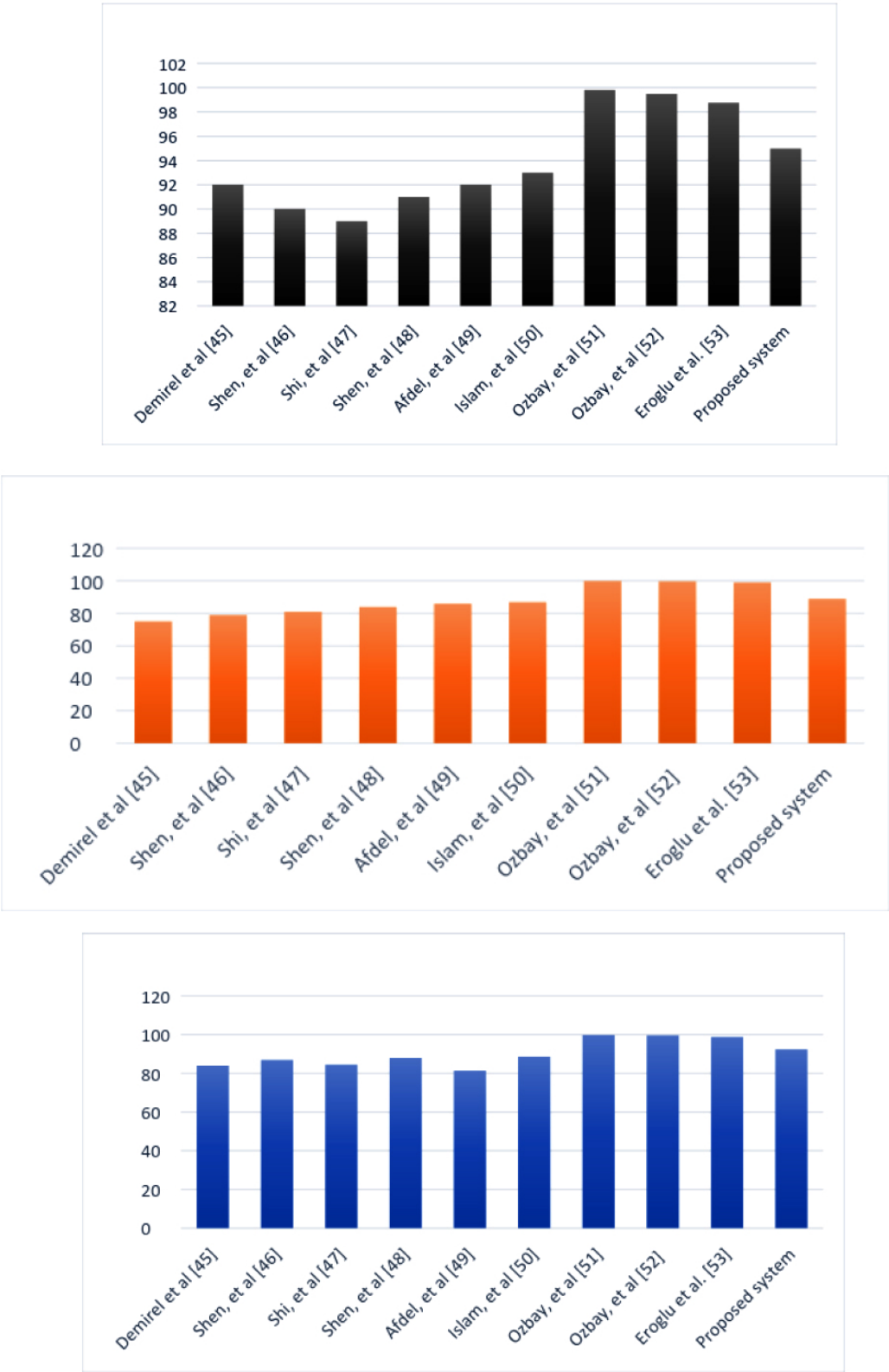
Our proposed framework has been compared to other existing techniques based on performance parameters such as accuracy, sensitivity, and specificity [45 - 50]. Fig. (11a, b, c) shows the graphical representation, while Table 3 shows the statistical representation. Table 6 depicts the overall analysis of various models over precision, recall, and F1-score. Fig. (12a, b, c) depicts a graphical representation of various models over precision, recall, and F1-score in which the proposed model outperforms another model.

**Table 5. Comparison of performance metrics.**

| Models                      | Accuracy | Sensitivity | Specificity |
|-----------------------------|----------|-------------|-------------|
| Demirel <i>et al</i> [45]   | 93.01    | 89.13       | 96.8        |
| Shen, <i>et al</i> [46]     | 91.02    | 92.7        | 89.05       |
| Shi, <i>et al</i> [47]      | 91.95    | 89.5        | 93.8        |
| Liu, <i>et al</i> [48]      | 92       | 90.9        | 93          |
| Aderghal, <i>et al</i> [49] | 91.41    | 89.06       | 93.75       |
| Islam, <i>et al</i> [50]    | 94.7     | 94.3        | 95.8        |
| Özbay <i>et al.</i> [51]    | 99.83    | 99.88       | 99.92       |
| Özbay <i>et al.</i> [52]    | 99.6     | 99.6        | 99.5        |
| Eroglu <i>et al.</i> [53]   | 98.75    | 98.83       | 99.32       |
| Proposed system             | 95.9     | 96.2        | 97.4        |

**Table 6. An overall analysis of various models over precision, recall, and F1-score.**

| Models                    | Precision | Recall | F1-score |
|---------------------------|-----------|--------|----------|
| Demirel <i>et al</i> [45] | 92        | 75     | 84       |
| Shen, <i>et al</i> [46]   | 90        | 79     | 87       |
| Shi, <i>et al</i> [47]    | 89        | 81     | 84.5     |
| Shen, <i>et al</i> [48]   | 91        | 84     | 88       |
| Afdel, <i>et al</i> [49]  | 92        | 86     | 81.3     |
| Islam, <i>et al</i> [50]  | 93        | 87     | 88.6     |
| Özbay <i>et al.</i> [51]  | 99.83     | 99.88  | 99.85    |
| Özbay <i>et al.</i> [52]  | 99.5      | 99.6   | 99.6     |
| Eroglu <i>et al.</i> [53] | 98.75     | 98.9   | 98.79    |
| Proposed system           | 95        | 89     | 92.4     |



**Fig. (12).** Models vs. measures a) precision, b) recall, c) F1-score.



Fig. (13). Models vs. measures a) detection rate, b) TPR, c) FPR.

Table 7 depicts the overall analysis of various models over detection rate, TPR, and FPR. Fig. (13a, b, c) depicts a

graphical representation of various models over detection rate, TPR, and FPR in which the proposed model outperforms other models.

**Table 7. An overall analysis of various models.**

| Models                    | Detection rate | TPR   | FPR |
|---------------------------|----------------|-------|-----|
| Demirel <i>et al</i> [45] | 91             | 86    | 14  |
| Shen, <i>et al</i> [46]   | 88             | 82    | 18  |
| Smith, <i>et al</i> [47]  | 86             | 80    | 20  |
| Liu, <i>et al</i> [48]    | 89             | 85    | 15  |
| Afdel, <i>et al</i> [49]  | 90             | 86    | 14  |
| Zhang, <i>et al</i> [50]  | 91             | 88    | 12  |
| Özbay <i>et al.</i> [51]  | 99.88          | 99.88 | 8   |
| Özbay <i>et al.</i> [52]  | 99.6           | 99.6  | 5   |
| Eroglu <i>et al.</i> [53] | 98.83          | 98.83 | 6   |
| Proposed system           | 94             | 91    | 9   |

## 5. DISCUSSION

The experimental studies presented in this section compare the performance of the proposed framework for Alzheimer's disease detection with several existing techniques. The comparison is based on various performance metrics, including accuracy, sensitivity, specificity, precision, recall, and F1-score.

Table 5 provides a comprehensive comparison of the performance metrics of different models, including Demirel *et al.* [45], Shen *et al.* [46], Shi *et al.* [47], Liu *et al.* [48], Aderghal *et al.* [49], Islam *et al.* [50], Özbay *et al.* [51], Özbay *et al.* [52], Eroglu *et al.* [53], and the proposed system. The proposed system achieves an accuracy of 95.9%, a sensitivity of 96.2%, and a specificity of 97.4%, outperforming most of the existing techniques. Özbay *et al.* [51, 52] exhibit higher accuracy, sensitivity, and specificity values, but it is important to note that these models achieve extremely high scores, which may indicate overfitting or potential issues with generalization to new data. Fig. (12) visualizes the performance metrics (accuracy, sensitivity, and specificity) of the different models, highlighting the superior performance of the proposed system. The figure clearly demonstrates that the proposed system consistently achieves higher accuracy, sensitivity, and specificity compared to the other models. Table 5 provides an overall analysis of the models in terms of precision, recall, and F1 score. The proposed system demonstrates a precision of 95%, a recall of 89%, and an F1-score of 92.4%. While some existing techniques show higher precision and recall values, the proposed system achieves a balanced performance with a competitive F1 score. Fig. (12) further illustrates the comparison of models based on precision, recall, and F1-score. The graph confirms that the proposed system performs favorably across these metrics, positioning it as an effective approach for Alzheimer's disease detection. In addition to the aforementioned metrics, Table 6 presents an analysis of the models' detection rate, true positive rate (TPR), and false positive rate (FPR). The proposed system achieves a detection rate of 94%, TPR of 91%, and FPR of 9%. These results indicate that the proposed system effectively detects Alzheimer's disease cases while maintaining a low false positive rate, which is crucial for minimizing misdiagnoses. Fig. (13) complements the analysis by visually comparing the models in terms of detection rate, TPR, and FPR. Once again, the proposed system outperforms the other models in terms of detection rate and TPR, while maintaining a competitive FPR.

Hence, the experimental studies demonstrate the superior

performance of the proposed framework for Alzheimer's disease detection. The system achieves high accuracy, sensitivity, specificity, precision, recall, and F1-score, while effectively detecting the disease and minimizing false positive rates. These findings support the effectiveness and potential clinical utility of the proposed framework in aiding the diagnosis of Alzheimer's disease. Further validation studies on larger and more diverse datasets would be valuable to consolidate these findings and assess the generalizability of the proposed approach.

## CONCLUSION

This research has presented a novel approach for the early detection and classification of Alzheimer's disease using a combination of image preprocessing techniques and deep learning models. The proposed framework leverages Adaptive Histogram Equalisation with boosted anisotropic diffusion filters for improved image quality and utilizes a modified ResNet-50 model with residual layer adjustments for feature extraction and classification.

Through extensive experimentation on the ADNI dataset, our proposed framework has achieved remarkable results, demonstrating its effectiveness in accurately identifying Alzheimer's disease. The obtained accuracy of 95% and minimum loss of 0.12 highlight the robustness and reliability of our approach. Furthermore, the comparative analysis against other state-of-the-art models has showcased the superior performance of our proposed method. The ResNet-50 architecture, with its intermediate depth and efficient feature extraction capabilities, strikes a balance between computational efficiency and improved performance, making it an optimal choice for practical applications where efficiency is crucial.

However, it is important to note that while our results are promising, there is still room for further improvement. Future work should focus on addressing the limitations of our method, such as potential overfitting and generalization issues, as well as exploring additional optimization techniques to enhance the overall performance. Therefore, this research contributes to the advancement of Alzheimer's disease diagnosis by providing a cutting-edge framework that combines innovative image preprocessing techniques and a tailored deep-learning architecture. The achieved results support the practical application of our model in various healthcare settings, with the potential to facilitate more reliable and timely diagnoses, ultimately leading to improved patient care and management.

## LIST OF ABBREVIATIONS

|                 |                                                 |
|-----------------|-------------------------------------------------|
| <b>AD</b>       | = Alzheimer's disease                           |
| <b>ResNet</b>   | = Deep Residual Network                         |
| <b>CLAHE</b>    | = Contrast Limited Adaptive Histogram Equalizer |
| <b>BADF</b>     | = Boosted Anisotropic Diffusion Filters         |
| <b>DL</b>       | = Deep learning                                 |
| <b>ML</b>       | = Machine learning                              |
| <b>DCNNs</b>    | = Deep-Convolutional Neural Networks            |
| <b>NCA</b>      | = Neighborhood Component Analysis               |
| <b>Grad-CAM</b> | = Gradient-weighted Class Activation Mapping    |
| <b>MCC</b>      | = Matthews Correlation Coefficient              |
| <b>BADF</b>     | = Boosted Anisotropic Diffusion Filter          |
| <b>ICP</b>      | = Iterative Closest Point                       |
| <b>CDF</b>      | = Cumulative Distribution Function              |
| <b>PDE</b>      | = Partial Differential Equation                 |
| <b>ADNI</b>     | = Alzheimer's Disease Neuroimaging Initiative   |
| <b>AD</b>       | = Alzheimer's disease                           |
| <b>SMOTE</b>    | = Synthetic Minority Oversampling Technique     |
| <b>TPR</b>      | = True positive rate                            |

## ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

## HUMAN AND ANIMAL RIGHTS

No animals/humans were used for studies that are the basis of this research.

## CONSENT FOR PUBLICATION

Not applicable.

## AVAILABILITY OF DATA AND MATERIALS

Not applicable.

## FUNDING

None.

## CONFLICT OF INTEREST

The authors declare no conflicts of interest, financial or otherwise.

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Declared none.

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